1 Preliminary Definitions

Symbol	Definition
$\mathbb{N}_0$	The set of non-negative integers, i.e., $\{0, 1, 2, 3, \dots\}$.
$\sum_{i=1}^{n} x_i$	The sum of the values $x_1, \dots, x_n$, i.e., $x_1 + \dots + x_n$.
$\prod_{i=1}^{n} x_i$	The product of the values $x_1, \dots, x_n$, i.e., $x_1 \cdot \dots \cdot x_n$.
A	A set of activity names, for example $A = \{a, b, c, d\}$.
A^*	The set of all sequences over the set A, for example, the set of sequences over $A = \{a, b, c, d\}$ is $A^* = \{\varepsilon, a, b, c, d, aa, ab, ac, ad, \dots\}$.
ε	The empty sequence of length 0.
σ	A trace, i.e., a sequence of activity names.
$\lvert\sigma\rvert$	The number of activities that the sequence σ consists of.
$\sigma(i)$	The i-th symbol in the sequence σ, where $i \in \{1, \dots, \lvert\sigma\rvert\}$.
L	An eventlog, i.e., a multiset of traces $L : A^* \to \mathbb{N}_0$.
$L(\sigma)$	The number of times the trace σ appears in L.
$\sigma \in L$	True if $L(\sigma) > 0$ is true.
$supp(L)$	The support of the event log, i.e., the set $\{\sigma \mid \sigma \in L\}$ of traces that appear at least once in L.
$A(\sigma)$	The set of activity names that appear in a trace σ, i.e., $A(\sigma) = \{\sigma(i) \mid i \in \{1, \dots, \lvert\sigma\rvert\}\}$.
$A(L)$	The set of activity names that appear in the event log L, i.e., $A(L) = \{\sigma(i) \mid \sigma \in L \wedge i \in \{1, \dots, \lvert\sigma\rvert\}\}$.
$A_I(L)$	The set of activity names that appear at the first position in a trace of L, i.e., $A_I(L) = \{\sigma(1) \mid \sigma \in L\}$
$A_O(L)$	The set of activity names that appear at the last position in a trace of L, i.e., $A_O(L) = \{\sigma(\lvert\sigma\rvert) \mid \sigma \in L\}$
$exp(x)$	Exponential function on the input x, i.e., e^x, where e is the Euler constant.
$\ln(x)$	The natural logarithm on the input x, i.e. the value y such that $x = e^y$.
$\lfloor r \rfloor$	The real number r, rounded down. For example, $\lfloor 3.6 \rfloor = 3$.
$\lceil r \rceil$	The real number r, rounded up. For example, $\lceil 3.2 \rceil = 4$.
$DFG(L)$	The directly follows graph $G = (V_{DFG}, E_{DFG})$ with $V_{DFG} = A(L)$ and $E_{DFG} = \{(a, b) \mid a \in V_{DFG} \wedge b \in V_{DFG} \wedge \exists \sigma \in L : \exists i \in \{1, \dots, \lvert\sigma\rvert - 1\} : \sigma(i) = a \wedge \sigma(i+1) = b\}$.

	Symbol	Description	Definition	Reference
1.	C_{len}	Number of traces in the log	$\sum_{\sigma \in L} L(\sigma)$	[1, 4, 6, 8]
2.	C_{var}	Number of distinct activity names in the log	$\lvert A(L) \rvert$	[1, 4, 6, 8]
3.	C_{mag}	Total number of events in the log	$\sum_{\sigma \in L} L(\sigma) \cdot \lvert \sigma \rvert$	[1, 4, 8]
4.	$C_{\text{A-start-\#}}$	Number of start activities in the log	$\lvert A_I(L) \rvert$	[1, 6, 8]
5.	$C_{\text{A-start-rel}}$	Relative number of start activities in the log	$\frac{C_{\text{A-start-\#}}(L)}{C_{\text{var}}(L)}$	[8]
6.	$C_{\text{A-start-min}}$	Minimum number of times a fixed activity occurs at the start of a trace	$\min_{a \in A_I(L)} \left\{ \sum_{\sigma \in L} L(\sigma) \cdot \lvert \{ i \in \{1\} \mid \sigma(i) = a \} \rvert \right\}$	[6]
7.	$C_{\text{A-start-max}}$	Maximum number of times a fixed activity occurs at the start of a trace	$\max_{a \in A_I(L)} \left\{ \sum_{\sigma \in L} L(\sigma) \cdot \lvert \{ i \in \{1\} \mid \sigma(i) = a \} \rvert \right\}$	[6]
8.	$C_{\text{A-start-mean}}$	Average number of times a fixed activity occurs at the start of a trace	$\frac{\sum_{a \in A_I(L)} \left\{ \sum_{\sigma \in L} L(\sigma) \cdot \lvert \{ i \in \{1\} \mid \sigma(i) = a \} \rvert \right\}}{C_{\text{var}}(L)}$	[6]
9.	$C_{\text{A-start-med}}$	The median number of times a fixed activity occurs at the start of a trace	$\begin{cases} \frac{1}{2} \cdot \left(a_{\frac{n}{2}} + a_{\frac{n}{2}+1} \right) & \text{if } n \text{ is even} \\ a_{\lceil \frac{n}{2} \rceil} & \text{otherwise} \end{cases}$ where $(a_1, \ldots, a_n)$ is a sorted list of $A_I(L)$'s occurrences at the start of a trace of L, in ascending order.	[6]
10.	$C_{\text{A-start-dev}}$	The standard deviation of times a fixed activity occurs at the start of a trace	$\sqrt{\frac{\sum_{a \in A_I(L)} \left(\left(\sum_{\sigma \in L} L(\sigma) \cdot \lvert \{ i \in \{1\} \mid \sigma(i) = a \} \rvert \right) - C_{\text{A-start-mean}}(L) \right)^2}{C_{\text{A-start-\#}}(L)}}$	[6]
11.	$C_{\text{A-start-var}}$	The variance of times a fixed activity occurs at the start of a trace	$\frac{\sum_{a \in A_I(L)} \left(\left(\sum_{\sigma \in L} L(\sigma) \cdot \lvert \{ i \in \{1\} \mid \sigma(i) = a \} \rvert \right) - C_{\text{A-start-mean}}(L) \right)^2}{C_{\text{A-start-\#}}(L)}$	[6]

	Symbol	Description	Definition	Reference
12.	$C_{\text{A-start-Q1}}$	The first quartile of times a fixed activity occurs at the start of a trace	$a_{\lfloor \frac{n}{4} \rfloor} + (\frac{n}{4} - \lfloor \frac{n}{4} \rfloor) \cdot (a_{\lfloor \frac{n}{4} \rfloor + 1} - a_{\lfloor \frac{n}{4} \rfloor})$, where $(a_1, \ldots, a_n)$ is a sorted list of $A_I(L)$'s occurrences at the start of a trace of L, in ascending order. If $n \in \{2, 3\}$, we take $a_1 + \frac{n}{4} \cdot (a_2 - a_1)$ instead. For $n = 1$, we take a_1.	[6]
13.	$C_{\text{A-start-Q3}}$	The third quartile of times a fixed activity occurs at the start of a trace	$a_{\lfloor \frac{3n}{4} \rfloor} + (\frac{3n}{4} - \lfloor \frac{3n}{4} \rfloor) \cdot (a_{\lfloor \frac{3n}{4} \rfloor + 1} - a_{\lfloor \frac{3n}{4} \rfloor})$, where $(a_1, \ldots, a_n)$ is a sorted list of $A_I(L)$'s occurrences at the start of a trace of L, in ascending order. If $n \in \{2, 3\}$, we take $a_1 + \frac{3n}{4} \cdot (a_2 - a_1)$ instead. For $n = 1$, we take a_1.	[6]
14.	$C_{\text{A-start-IQR}}$	The interquartile range of times a fixed activity occurs at the start of a trace	$C_{\text{A-start-Q3}}(L) - C_{\text{A-start-Q1}}(L)$	[6]
15.	$C_{\text{A-start-skew}}$	The skewness of times a fixed activity occurs at the start of a trace	$\frac{\frac{1}{C_{\text{A-start-\#}}(L)} \sum_{a \in A_I(L)} \left(\left(\sum_{\sigma \in L} L(\sigma) \cdot \lvert \{i \in \{1\} \mid \sigma(i) = a\} \rvert \right) - C_{\text{A-start-mean}}(L) \right)^3}{\left(\frac{1}{C_{\text{A-start-\#}}(L)} \sum_{a \in A_I(L)} \left(\left(\sum_{\sigma \in L} L(\sigma) \cdot \lvert \{i \in \{1\} \mid \sigma(i) = a\} \rvert \right) - C_{\text{A-start-mean}}(L) \right)^2 \right)^{1.5}}$	[6]
16.	$C_{\text{A-start-kurt}}$	The kurtosis of times a fixed activity occurs at the start of a trace	$\frac{\frac{1}{C_{\text{A-start-\#}}(L)} \sum_{a \in A_I(L)} \left(\left(\sum_{\sigma \in L} L(\sigma) \cdot \lvert \{i \in \{1\} \mid \sigma(i) = a\} \rvert \right) - C_{\text{A-start-mean}}(L) \right)^4}{\left(\frac{1}{C_{\text{A-start-\#}}(L)} \sum_{a \in A_I(L)} \left(\left(\sum_{\sigma \in L} L(\sigma) \cdot \lvert \{i \in \{1\} \mid \sigma(i) = a\} \rvert \right) - C_{\text{A-start-mean}}(L) \right)^2 \right)^2} - 3$	[6]
17.	$C_{\text{A-end-\#}}$	Number of end activities in the log	$\lvert A_O(L) \rvert$	[1, 6, 8]
18.	$C_{\text{A-end-rel}}$	Relative number of end activities in the log	$\frac{C_{\text{A-end-\#}}(L)}{C_{\text{var}}(L)}$	[8]
19.	$C_{\text{A-end-min}}$	Minimum number of times a fixed activity occurs at the end of a trace	$\min_{a \in A_O(L)} \left\{ \sum_{\sigma \in L} L(\sigma) \cdot \lvert \{i \in \{\lvert \sigma \rvert\} \mid \sigma(i) = a\} \rvert \right\}$	[6]
20.	$C_{\text{A-end-max}}$	Maximum number of times a fixed activity occurs at the end of a trace	$\max_{a \in A_O(L)} \left\{ \sum_{\sigma \in L} L(\sigma) \cdot \lvert \{i \in \{\lvert \sigma \rvert\} \mid \sigma(i) = a\} \rvert \right\}$	[6]
21.	$C_{\text{A-end-mean}}$	Average number of times a fixed activity occurs at the end of a trace	$\frac{\sum_{a \in A_O(L)} \left\{ \sum_{\sigma \in L} L(\sigma) \cdot \lvert \{i \in \{\lvert \sigma \rvert\} \mid \sigma(i) = a\} \rvert \right\}}{C_{\text{var}}(L)}$	[6]

	Symbol	Description	Definition	Reference
22.	$C_{\text{A-end-med}}$	The median number of times a fixed activity occurs at the end of a trace	$\begin{cases} \frac{1}{2} \cdot \left(a_{\frac{n}{2}} + a_{\frac{n}{2}+1}\right) & \text{if } n \text{ is even} \\ a_{\lceil \frac{n}{2} \rceil} & \text{otherwise} \end{cases}$ where $(a_1, \ldots, a_n)$ is a sorted list of $A_O(L)$'s occurrences at the end of a trace of L, in ascending order.	[6]
23.	$C_{\text{A-end-dev}}$	The standard deviation of times a fixed activity occurs at the end of a trace	$\sqrt{\frac{\sum_{a \in A_O(L)} \left(\left(\sum_{\sigma \in L} L(\sigma) \cdot \lvert\{i \in \{\lvert\sigma\rvert\} \mid \sigma(i) = a\}\rvert\right) - C_{\text{A-start-mean}}(L)\right)^2}{C_{\text{A-start-\#}}(L)}}$	[6]
24.	$C_{\text{A-end-var}}$	The variance of times a fixed activity occurs at the end of a trace	$\frac{\sum_{a \in A_O(L)} \left(\left(\sum_{\sigma \in L} L(\sigma) \cdot \lvert\{i \in \{\lvert\sigma\rvert\} \mid \sigma(i) = a\}\rvert\right) - C_{\text{A-start-mean}}(L)\right)^2}{C_{\text{A-start-\#}}(L)}$	[6]
25.	$C_{\text{A-end-Q1}}$	The first quartile of times a fixed activity occurs at the end of a trace	$a_{\lfloor \frac{n}{4} \rfloor} + (\frac{n}{4} - \lfloor \frac{n}{4} \rfloor) \cdot (a_{\lfloor \frac{n}{4} \rfloor + 1} - a_{\lfloor \frac{n}{4} \rfloor})$, where $(a_1, \ldots, a_n)$ is a sorted list of $A_O(L)$'s occurrences at the end of a trace of L, in ascending order. If $n \in \{2, 3\}$, we take $a_1 + \frac{n}{4} \cdot (a_2 - a_1)$ instead. For $n = 1$, we take a_1.	[6]
26.	$C_{\text{A-end-Q3}}$	The third quartile of times a fixed activity occurs at the end of a trace	$a_{\lfloor \frac{3n}{4} \rfloor} + (\frac{3n}{4} - \lfloor \frac{3n}{4} \rfloor) \cdot (a_{\lfloor \frac{3n}{4} \rfloor + 1} - a_{\lfloor \frac{3n}{4} \rfloor})$, where $(a_1, \ldots, a_n)$ is a sorted list of $A_O(L)$'s occurrences at the end of a trace of L, in ascending order. If $n \in \{2, 3\}$, we take $a_1 + \frac{3n}{4} \cdot (a_2 - a_1)$ instead. For $n = 1$, we take a_1.	[6]
27.	$C_{\text{A-end-IQR}}$	The interquartile range of times a fixed activity occurs at the end of a trace	$C_{\text{A-end-Q3}}(L) - C_{\text{A-end-Q1}}(L)$	[6]
28.	$C_{\text{A-end-skew}}$	The skewness of times a fixed activity occurs at the end of a trace	$\frac{\frac{1}{C_{\text{A-end-\#}}(L)} \sum_{a \in A_O(L)} \left(\left(\sum_{\sigma \in L} L(\sigma) \cdot \lvert\{i \in \{\lvert\sigma\rvert\} \mid \sigma(i) = a\}\rvert\right) - C_{\text{A-end-mean}}(L)\right)^3}{\left(\frac{1}{C_{\text{A-end-\#}}(L)} \sum_{a \in A_O(L)} \left(\left(\sum_{\sigma \in L} L(\sigma) \cdot \lvert\{i \in \{\lvert\sigma\rvert\} \mid \sigma(i) = a\}\rvert\right) - C_{\text{A-end-mean}}(L)\right)^2\right)^{1.5}}$	[6]
29.	$C_{\text{A-end-kurt}}$	The kurtosis of times a fixed activity occurs at the end of a trace	$\frac{\frac{1}{C_{\text{A-end-\#}}(L)} \sum_{a \in A_O(L)} \left(\left(\sum_{\sigma \in L} L(\sigma) \cdot \lvert\{i \in \{\lvert\sigma\rvert\} \mid \sigma(i) = a\}\rvert\right) - C_{\text{A-end-mean}}(L)\right)^4}{\left(\frac{1}{C_{\text{A-end-\#}}(L)} \sum_{a \in A_O(L)} \left(\left(\sum_{\sigma \in L} L(\sigma) \cdot \lvert\{i \in \{\lvert\sigma\rvert\} \mid \sigma(i) = a\}\rvert\right) - C_{\text{A-end-mean}}(L)\right)^2\right)^2} - 3$	[6]
30.	$C_{\text{TL-min}}$	Minimum length of a trace in the log	$\min\{\lvert\sigma\rvert \mid \sigma \in L\}$	[1, 6, 8]
31.	$C_{\text{TL-max}}$	Maximum length of a trace in the log	$\max\{\lvert\sigma\rvert \mid \sigma \in L\}$	[1, 6, 8]

	Symbol	Description	Definition	Reference
32.	$C_{\text{TL-mean}}$	Average length of the traces in the log	$\frac{\sum_{\sigma \in L} L(\sigma) \cdot \lvert\sigma\rvert}{\sum_{\sigma \in L} L(\sigma)}$	[1, 8]
33.	$C_{\text{TL-med}}$	The median of all trace lengths in the log	$\begin{cases} \frac{1}{2} \cdot \left(\ell_{\frac{n}{2}} + \ell_{\frac{n}{2}+1}\right) & \text{if } n \text{ is even} \\ \ell_{\lceil \frac{n}{2} \rceil} & \text{otherwise} \end{cases}$ where $(\ell_1, \ldots, \ell_n)$ is a list of L's trace lengths, in ascending order.	[6]
34.	$C_{\text{TL-mode}}$	A most frequent length of a trace in the log	some element of $\{\lvert\sigma\rvert \mid \forall \sigma' \in L : \lvert L^{\lvert\sigma\rvert}\rvert \geq \lvert L^{\lvert\sigma'\rvert}\rvert\}$ where we set $L^n = \{\sigma \in L \mid \lvert\sigma\rvert = n\}$.	[6]
35.	$C_{\text{TL-dev}}$	Standard deviation of trace lengths in the log	$\sqrt{\frac{\sum_{\sigma \in L} L(\sigma) \cdot (\lvert\sigma\rvert - C_{\text{TL-mean}}(L))^2}{C_{\text{len}}(L)}}$	[1, 6]
36.	$C_{\text{TL-var}}$	Variance of trace lengths in the log	$\frac{\sum_{\sigma \in L} L(\sigma) \cdot (\lvert\sigma\rvert - C_{\text{TL-mean}}(L))^2}{C_{\text{len}}(L)}$	[6]
37.	$C_{\text{TL-Q1}}$	The first quartile of trace lengths in the log	$\ell_{\lfloor \frac{n}{4} \rfloor} + (\frac{n}{4} - \lfloor \frac{n}{4} \rfloor) \cdot (\ell_{\lfloor \frac{n}{4} \rfloor + 1} - \ell_{\lfloor \frac{n}{4} \rfloor})$, where $(\ell_1, \ldots, \ell_n)$ is a sorted list of L's trace lengths, in ascending order. If $n \in \{2, 3\}$, we take $\ell_1 + \frac{n}{4} \cdot (\ell_2 - \ell_1)$ instead. For $n = 1$, we take ℓ_1.	[6]
38.	$C_{\text{TL-Q3}}$	The third quartile of trace lengths in the log	$\ell_{\lfloor \frac{3n}{4} \rfloor} + (\frac{3n}{4} - \lfloor \frac{3n}{4} \rfloor) \cdot (\ell_{\lfloor \frac{3n}{4} \rfloor + 1} - \ell_{\lfloor \frac{3n}{4} \rfloor})$, where $(\ell_1, \ldots, \ell_n)$ is a sorted list of L's trace lengths, in ascending order. If $n \in \{2, 3\}$, we take $\ell_1 + \frac{3n}{4} \cdot (\ell_2 - \ell_1)$ instead. For $n = 1$, we take ℓ_1.	[6]
39.	$C_{\text{TL-IQR}}$	The interquartile range of trace lenghts in the log	$C_{\text{TL-Q3}}(L) - C_{\text{TL-Q1}}(L)$	[6]
40.	$C_{\text{TL-g-mean}}$	The geometric mean of the log's trace lengths	$\left(\prod_{\sigma \in L} \lvert\sigma\rvert^{L(\sigma)}\right)^{\frac{1}{C_{\text{len}}(L)}}$	[6]
41.	$C_{\text{TL-g-dev}}$	The geometric standard deviation of the log's trace lengths	$exp\left(\sqrt{\frac{1}{C_{\text{len}}(L)} \sum_{\sigma \in L} L(\sigma) \cdot \left(\ln\left(\frac{\lvert\sigma\rvert}{C_{\text{TL-g-mean}}(L)}\right)\right)^2}\right)$	[6]
42.	$C_{\text{TL-h-mean}}$	The harmonic mean of the log's trace lengths	$\frac{C_{\text{len}}(L)}{\sum_{\sigma \in L} \frac{1}{\lvert\sigma\rvert}}$	[6]

	Symbol	Description	Definition	Reference
43.	$C_{\text{TL-skew}}$	The skewness of the log's trace lengths	$\frac{\frac{1}{C_{\text{len}(L)}} \sum_{\sigma \in L} L(\sigma) \cdot (\lvert\sigma\rvert - C_{\text{TL-mean}}(L))^3}{\left(\frac{1}{C_{\text{len}(L)}} \sum_{\sigma \in L} L(\sigma) \cdot (\lvert\sigma\rvert - C_{\text{TL-mean}}(L))^2\right)^{1.5}}$	[6]
44.	$C_{\text{TL-kurt}}$	The kurtosis of the log's trace lengths (Fisher's definition)	$\frac{\frac{1}{C_{\text{len}(L)}} \sum_{\sigma \in L} L(\sigma) \cdot (\lvert\sigma\rvert - C_{\text{TL-mean}}(L))^4}{\left(\frac{1}{C_{\text{len}(L)}} \sum_{\sigma \in L} L(\sigma) \cdot (\lvert\sigma\rvert - C_{\text{TL-mean}}(L))^2\right)^2} - 3$	[6]
45.	$C_{\text{TL-cvar}}$	The coefficient variation of the log's trace lengths	$\frac{C_{\text{TL-dev}}(L)}{C_{\text{TL-mean}}(L)}$	[6]
46.	$C_{\text{TL-ent}}$	The Shannon entropy of the log's trace lengths	$-\sum_{\sigma \in L} \left(\frac{L(\sigma) \cdot \lvert\sigma\rvert}{C_{\text{mag}}(L)} \cdot \ln(\frac{L(\sigma) \cdot \lvert\sigma\rvert}{C_{\text{mag}}(L)} \right)$	[6]
47.	$C_{\text{TL-hist1}}$	The first value of the trace lengths' histogram	$\frac{\sum_{\sigma \in \mathcal{B}_1} L(\sigma)}{\frac{1}{10} \cdot C_{\text{len}}(L) \cdot (\ell_{\max} - \ell_{\min})}$ where $\ell_{\min} = \min\{\lvert\sigma\rvert \mid \sigma \in L\}$, $\ell_{\max} = \max\{\lvert\sigma\rvert \mid \sigma \in L\}$, and $\mathcal{B}_1 = \{\sigma \mid \ell_{\min} \leq \lvert\sigma\rvert < \ell_{\min} + \frac{\ell_{max} - \ell_{min}}{10}\}$	[6]
48.	$C_{\text{TL-hist2}}$	The second value of the trace lengths' histogram	$\frac{\sum_{\sigma \in \mathcal{B}_2} L(\sigma)}{\frac{1}{10} \cdot C_{\text{len}}(L) \cdot (\ell_{\max} - \ell_{\min})}$ where $\ell_{\min} = \min\{\lvert\sigma\rvert \mid \sigma \in L\}$, $\ell_{\max} = \max\{\lvert\sigma\rvert \mid \sigma \in L\}$, and $\mathcal{B}_2 = \{\sigma \mid \ell_{\min} + \frac{\ell_{max} - \ell_{min}}{10} \leq \lvert\sigma\rvert < \ell_{\min} + 2 \cdot \frac{\ell_{max} - \ell_{min}}{10}\}$	[6]
49.	$C_{\text{TL-hist3}}$	The third value of the trace lengths' histogram	$\frac{\sum_{\sigma \in \mathcal{B}_3} L(\sigma)}{\frac{1}{10} \cdot C_{\text{len}}(L) \cdot (\ell_{\max} - \ell_{\min})}$ where $\ell_{\min} = \min\{\lvert\sigma\rvert \mid \sigma \in L\}$, $\ell_{\max} = \max\{\lvert\sigma\rvert \mid \sigma \in L\}$, and $\mathcal{B}_3 = \{\sigma \mid \ell_{\min} + 2 \cdot \frac{\ell_{max} - \ell_{min}}{10} \leq \lvert\sigma\rvert < \ell_{\min} + 3 \cdot \frac{\ell_{max} - \ell_{min}}{10}\}$	[6]
50.	$C_{\text{TL-hist4}}$	The fourth value of the trace lengths' histogram	$\frac{\sum_{\sigma \in \mathcal{B}_4} L(\sigma)}{\frac{1}{10} \cdot C_{\text{len}}(L) \cdot (\ell_{\max} - \ell_{\min})}$ where $\ell_{\min} = \min\{\lvert\sigma\rvert \mid \sigma \in L\}$, $\ell_{\max} = \max\{\lvert\sigma\rvert \mid \sigma \in L\}$, and $\mathcal{B}_4 = \{\sigma \mid \ell_{\min} + 3 \cdot \frac{\ell_{max} - \ell_{min}}{10} \leq \lvert\sigma\rvert < \ell_{\min} + 4 \cdot \frac{\ell_{max} - \ell_{min}}{10}\}$	[6]
51.	$C_{\text{TL-hist5}}$	The fifth value of the trace lengths' histogram	$\frac{\sum_{\sigma \in \mathcal{B}_5} L(\sigma)}{\frac{1}{10} \cdot C_{\text{len}}(L) \cdot (\ell_{\max} - \ell_{\min})}$ where $\ell_{\min} = \min\{\lvert\sigma\rvert \mid \sigma \in L\}$, $\ell_{\max} = \max\{\lvert\sigma\rvert \mid \sigma \in L\}$, and $\mathcal{B}_5 = \{\sigma \mid \ell_{\min} + 4 \cdot \frac{\ell_{max} - \ell_{min}}{10} \leq \lvert\sigma\rvert < \ell_{\min} + 5 \cdot \frac{\ell_{max} - \ell_{min}}{10}\}$	[6]

	Symbol	Description	Definition	Reference
52.	$C_{\text{TL-hist6}}$	The sixth value of the trace lengths' histogram	$\frac{\sum_{\sigma \in \mathcal{B}_6} L(\sigma)}{\frac{1}{10} \cdot C_{\text{len}}(L) \cdot (\ell_{\max} - \ell_{\min})}$ where $\ell_{\min} = \min\{\lvert\sigma\rvert \mid \sigma \in L\}$, $\ell_{\max} = \max\{\lvert\sigma\rvert \mid \sigma \in L\}$, and $\mathcal{B}_6 = \{\sigma \mid \ell_{\min} + 5 \cdot \frac{\ell_{max} - \ell_{min}}{10} \leq \lvert\sigma\rvert < \ell_{\min} + 6 \cdot \frac{\ell_{max} - \ell_{min}}{10}\}$	[6]
53.	$C_{\text{TL-hist7}}$	The seventh value of the trace lengths' histogram	$\frac{\sum_{\sigma \in \mathcal{B}_7} L(\sigma)}{\frac{1}{10} \cdot C_{\text{len}}(L) \cdot (\ell_{\max} - \ell_{\min})}$ where $\ell_{\min} = \min\{\lvert\sigma\rvert \mid \sigma \in L\}$, $\ell_{\max} = \max\{\lvert\sigma\rvert \mid \sigma \in L\}$, and $\mathcal{B}_7 = \{\sigma \mid \ell_{\min} + 6 \cdot \frac{\ell_{max} - \ell_{min}}{10} \leq \lvert\sigma\rvert < \ell_{\min} + 7 \cdot \frac{\ell_{max} - \ell_{min}}{10}\}$	[6]
54.	$C_{\text{TL-hist8}}$	The eighth value of the trace lengths' histogram	$\frac{\sum_{\sigma \in \mathcal{B}_8} L(\sigma)}{\frac{1}{10} \cdot C_{\text{len}}(L) \cdot (\ell_{\max} - \ell_{\min})}$ where $\ell_{\min} = \min\{\lvert\sigma\rvert \mid \sigma \in L\}$, $\ell_{\max} = \max\{\lvert\sigma\rvert \mid \sigma \in L\}$, and $\mathcal{B}_8 = \{\sigma \mid \ell_{\min} + 7 \cdot \frac{\ell_{max} - \ell_{min}}{10} \leq \lvert\sigma\rvert < \ell_{\min} + 8 \cdot \frac{\ell_{max} - \ell_{min}}{10}\}$	[6]
55.	$C_{\text{TL-hist9}}$	The ninth value of the trace lengths' histogram	$\frac{\sum_{\sigma \in \mathcal{B}_9} L(\sigma)}{\frac{1}{10} \cdot C_{\text{len}}(L) \cdot (\ell_{\max} - \ell_{\min})}$ where $\ell_{\min} = \min\{\lvert\sigma\rvert \mid \sigma \in L\}$, $\ell_{\max} = \max\{\lvert\sigma\rvert \mid \sigma \in L\}$, and $\mathcal{B}_9 = \{\sigma \mid \ell_{\min} + 8 \cdot \frac{\ell_{max} - \ell_{min}}{10} \leq \lvert\sigma\rvert < \ell_{\min} + 9 \cdot \frac{\ell_{max} - \ell_{min}}{10}\}$	[6]
56.	$C_{\text{TL-hist10}}$	The tenth value of the trace lengths' histogram	$\frac{\sum_{\sigma \in \mathcal{B}_{10}} L(\sigma)}{\frac{1}{10} \cdot C_{\text{len}}(L) \cdot (\ell_{\max} - \ell_{\min})}$ where $\ell_{\min} = \min\{\lvert\sigma\rvert \mid \sigma \in L\}$, $\ell_{\max} = \max\{\lvert\sigma\rvert \mid \sigma \in L\}$, and $\mathcal{B}_{10} = \{\sigma \mid \ell_{\min} + 9 \cdot \frac{\ell_{max} - \ell_{min}}{10} \leq \lvert\sigma\rvert \leq \ell_{\min} + 10 \cdot \frac{\ell_{max} - \ell_{min}}{10}\}$	[6]
57.	$C_{\text{TL-hist-skew}}$	The skewness of the trace lengths' histogram	$\frac{\frac{1}{10} \sum_{i=1}^{10} (C_{\text{hist}i}(L) - \mu_{\text{hist}}(L))^3}{\left(\frac{1}{10} \sum_{i=1}^{10} (C_{\text{hist}i}(L) - \mu_{\text{hist}}(L))^2\right)^{1.5}}$ where $\mu_{\text{hist}}(L) = \frac{\sum_{i=1}^{10} C_{\text{hist}i}(L)}{10}$	[6]
58.	$C_{\text{TL-hist-kurt}}$	The kurtosis of the trace lengths' histogram (Fisher's definition)	$\frac{\frac{1}{10} \sum_{i=1}^{10} (C_{\text{hist}i}(L) - \mu_{\text{hist}}(L))^4}{\left(\frac{1}{10} \sum_{i=1}^{10} (C_{\text{hist}i}(L) - \mu_{\text{hist}}(L))^2\right)^2} - 3$ where $\mu_{\text{hist}}(L) = \frac{\sum_{i=1}^{10} C_{\text{hist}i}(L)}{10}$	[6]
59.	$C_{\text{FREQ-T-max-r}}$	The highest frequency in the log divided by the total number of traces in the log	$\frac{\max\{L(\sigma) \mid \sigma \in L\}}{C_{\text{len}}(L)}$	[6]

	Symbol	Description	Definition	Reference
60.	$C_{\text{FREQ-T-max-1\%}}$	Ratio of the 1% highest frequencies in the log	$\frac{\sum_{i=1}^{\lfloor 0.01 \cdot C_{\text{len}}(L) \rfloor} L(\omega_i)}{C_{\text{len}}(L)}$, where $[\omega_1^{L(\omega_1)}, \ldots, \omega_n^{L(\omega_n)}]$ is the event log L sorted in descending order by frequency.	[6]
61.	$C_{\text{FREQ-T-max-5\%}}$	Ratio of the 5% highest frequencies in the log	$\frac{\sum_{i=1}^{\lfloor 0.05 \cdot C_{\text{len}}(L) \rfloor} L(\omega_i)}{C_{\text{len}}(L)}$, where $[\omega_1^{L(\omega_1)}, \ldots, \omega_n^{L(\omega_n)}]$ is the event log L sorted in descending order by frequency.	[6]
62.	$C_{\text{FREQ-T-max-10\%}}$	Ratio of the 10% highest frequencies in the log	$\frac{\sum_{i=1}^{\lfloor 0.1 \cdot C_{\text{len}}(L) \rfloor} L(\omega_i)}{C_{\text{len}}(L)}$, where $[\omega_1^{L(\omega_1)}, \ldots, \omega_n^{L(\omega_n)}]$ is the event log L sorted in descending order by frequency.	[6]
63.	$C_{\text{FREQ-T-max-20\%}}$	Ratio of the 20% highest frequencies in the log	$\frac{\sum_{i=1}^{\lfloor 0.2 \cdot C_{\text{len}}(L) \rfloor} L(\omega_i)}{C_{\text{len}}(L)}$, where $[\omega_1^{L(\omega_1)}, \ldots, \omega_n^{L(\omega_n)}]$ is the event log L sorted in descending order by frequency.	[6]
64.	$C_{\text{FREQ-T-max-50\%}}$	Ratio of the 50% highest frequencies in the log	$\frac{\sum_{i=1}^{\lfloor 0.5 \cdot C_{\text{len}}(L) \rfloor} L(\omega_i)}{C_{\text{len}}(L)}$, where $[\omega_1^{L(\omega_1)}, \ldots, \omega_n^{L(\omega_n)}]$ is the event log L sorted in descending order by frequency.	[6]
65.	$C_{\text{FREQ-T-max-75\%}}$	Ratio of the 75% highest frequencies in the log	$\frac{\sum_{i=1}^{\lfloor 0.75 \cdot C_{\text{len}}(L) \rfloor} L(\omega_i)}{C_{\text{len}}(L)}$, where $[\omega_1^{L(\omega_1)}, \ldots, \omega_n^{L(\omega_n)}]$ is the event log L sorted in descending order by frequency.	[6]
66.	$C_{\text{FREQ-T-mean}}$	Average frequencies of traces in the event log	$\frac{\sum_{\sigma \in L} L(\sigma)}{\lvert supp(L) \rvert}$	[6]
67.	$C_{\text{FREQ-T-dev}}$	Standard deviation of frequencies of traces in the event log	$\sqrt{\frac{\sum_{\sigma \in L} (L(\sigma) - C_{\text{FREQ-T-mean}}(L))^2}{\lvert supp(L) \rvert}}$	[6]
68.	$C_{\text{FREQ-T-skew}}$	The skewness of the frequencies of traces in the event log	$\frac{\frac{1}{\lvert supp(L) \rvert} \sum_{\sigma \in L} (L(\sigma) - C_{\text{FREQ-T-mean}}(L))^3}{\left(\frac{1}{\lvert supp(L) \rvert} \sum_{\sigma \in L} (L(\sigma) - C_{\text{FREQ-T-mean}}(L))^2\right)^{1.5}}$	[6]
69.	$C_{\text{FREQ-T-kurt}}$	The kurtosis of the frequencies of traces in the event log	$\frac{\frac{1}{\lvert supp(L) \rvert} \sum_{\sigma \in L} (L(\sigma) - C_{\text{FREQ-T-mean}}(L))^4}{\left(\frac{1}{\lvert supp(L) \rvert} \sum_{\sigma \in L} (L(\sigma) - C_{\text{FREQ-T-mean}}(L))^2\right)^2} - 3$	[6]

	Symbol	Description	Definition	Reference
70.	$C_{\text{FREQ-A-min}}$	The frequency of an event that occurs least often in the event log	$\min_{a \in A(L)} \left\{ \sum_{\sigma \in L} L(\sigma) \cdot \lvert \{ i \in \{1, \ldots, \lvert\sigma\rvert\} \mid \sigma(i) = a \} \rvert \right\}$	[6]
71.	$C_{\text{FREQ-A-max}}$	The frequency of an event that occurs most often in the event log	$\max_{a \in A(L)} \left\{ \sum_{\sigma \in L} L(\sigma) \cdot \lvert \{ i \in \{1, \ldots, \lvert\sigma\rvert\} \mid \sigma(i) = a \} \rvert \right\}$	[6]
72.	$C_{\text{FREQ-A-mean}}$	The average occurrence of an activity in the log	$\frac{\sum_{a \in A(L)} \sum_{\sigma \in L} L(\sigma) \cdot \lvert \{ i \in \{1, \ldots, \lvert\sigma\rvert\} \rvert \sigma(i) = a \} \rvert}{C_{\text{var}}(L)}$	[6]
73.	$C_{\text{FREQ-A-med}}$	The median of all occurrences of an activity in the log	$\begin{cases} \frac{1}{2} \cdot \left(a_{\frac{n}{2}} + a_{\frac{n}{2}+1} \right) & \text{if } n \text{ is even} \\ a_{\lceil \frac{n}{2} \rceil} & \text{otherwise} \end{cases}$ where $(a_1, \ldots, a_n)$ is a sorted list of $A(L)$'s occurrences, in ascending order.	[6]
74.	$C_{\text{FREQ-A-dev}}$	Standard deviation of frequencies of activity names in the event log	$\sqrt{\frac{\sum_{a \in A} \left(\left(\sum_{\sigma \in L} L(\sigma) \cdot \lvert \{ i \in \{1, \ldots, \lvert\sigma\rvert\} \rvert \sigma(i) = a \} \rvert \right) - C_{\text{FREQ-A-mean}}(L) \right)^2}{C_{\text{var}}(L)}}$	[6]
75.	$C_{\text{FREQ-A-var}}$	Variance of frequencies of activity names in the log	$\frac{\sum_{a \in A} \left(\left(\sum_{\sigma \in L} L(\sigma) \cdot \lvert \{ i \in \{1, \ldots, \lvert\sigma\rvert\} \rvert \sigma(i) = a \} \rvert \right) - C_{\text{FREQ-A-mean}}(L) \right)^2}{C_{\text{var}}(L)}$	[6]
76.	$C_{\text{FREQ-A-Q1}}$	The first quartile of times an activity occurs in a trace of the event log	$a_{\lfloor \frac{n}{4} \rfloor} + (\frac{n}{4} - \lfloor \frac{n}{4} \rfloor) \cdot (a_{\lfloor \frac{n}{4} \rfloor + 1} - a_{\lfloor \frac{n}{4} \rfloor})$, where $(a_1, \ldots, a_n)$ is a sorted list of $A(L)$'s occurrences, in ascending order. If $n \in \{2, 3\}$, we take $a_1 + \frac{n}{4} \cdot (a_2 - a_1)$ instead. For $n = 1$, we take a_1.	[6]
77.	$C_{\text{FREQ-A-Q3}}$	The third quartile of times an activity occurs in a trace of the event log	$a_{\lfloor \frac{3n}{4} \rfloor} + (\frac{3n}{4} - \lfloor \frac{3n}{4} \rfloor) \cdot (a_{\lfloor \frac{3n}{4} \rfloor + 1} - a_{\lfloor \frac{3n}{4} \rfloor})$, where $(a_1, \ldots, a_n)$ is a sorted list of $A(L)$'s occurrences, in ascending order. If $n \in \{2, 3\}$, we take $a_1 + \frac{3n}{4} \cdot (a_2 - a_1)$ instead. For $n = 1$, we take a_1.	[6]
78.	$C_{\text{FREQ-A-IRQ}}$	The interquartile range of times an activity occurs in a trace of the event log	$C_{\text{FREQ-A-Q3}}(L) - C_{\text{FREQ-A-Q1}}(L)$	[6]

	Symbol	Description	Definition	Reference
79.	$C_{\text{FREQ-A-skew}}$	The skewness of the activities' occurrences in the log	$\frac{\frac{1}{C_{\text{var}}(L)} \sum_{a \in A} \left(\left(\sum_{\sigma \in L} L(\sigma) \cdot \lvert \{ i \in \{1,\ldots,\lvert\sigma\rvert\} \rvert \sigma(i) = a \} \rvert \right) - C_{\text{FREQ-A-mean}}(L) \right)^3}{\left(\frac{1}{C_{\text{var}}(L)} \sum_{a \in A} \left(\left(\sum_{\sigma \in L} L(\sigma) \cdot \lvert \{ i \in \{1,\ldots,\lvert\sigma\rvert\} \rvert \sigma(i) = a \} \rvert \right) - C_{\text{FREQ-A-mean}}(L) \right)^2 \right)^{1.5}}$	[6]
80.	$C_{\text{FREQ-A-kurt}}$	The kurtosis of the activities' occurrences in the log	$\frac{\frac{1}{C_{\text{var}}(L)} \sum_{a \in A} \left(\left(\sum_{\sigma \in L} L(\sigma) \cdot \lvert \{ i \in \{1,\ldots,\lvert\sigma\rvert\} \rvert \sigma(i) = a \} \rvert \right) - C_{\text{FREQ-A-mean}}(L) \right)^4}{\left(\frac{1}{C_{\text{var}}(L)} \sum_{a \in A} \left(\left(\sum_{\sigma \in L} L(\sigma) \cdot \lvert \{ i \in \{1,\ldots,\lvert\sigma\rvert\} \rvert \sigma(i) = a \} \rvert \right) - C_{\text{FREQ-A-mean}}(L) \right)^2 \right)^2}$	[6]
81.	$C_{\text{DT-\#}}$	Number of distinct traces in the event log	$\lvert supp(L) \rvert$	[1, 6, 8]
82.	$C_{\text{DT-\%}}$	Ratio of distinct traces in the event log	$\frac{C_{\text{DT-\#}}(L)}{C_{\text{len}}(L)}$	[6, 8]
83.	$C_{\text{DT-cov}}$	The absolute trace coverage	$\min\{k \in \{0,\ldots,n\} \mid \sum_{i=1}^{k} f_i \geq 0.8 \cdot C_{\text{len}}(L)\}$ where $(f_1,\ldots,f_n)$ is a sorted list of the frequencies of traces in L, in ascending order.	[8]
84.	$C_{\text{DT-cov-\%}}$	The relative trace coverage	$\frac{C_{\text{DT-cov}}(L)}{C_{\text{DT-\#}}(L)}$	[8]
85.	$C_{\text{DT-het}}$	The heterogeneity rate of the traces in the log	$\frac{\ln(C_{\text{DT-\#}}(L))}{\ln(C_{\text{len}}(L))}$	[8]
86.	$C_{\text{DT-sim}}$	The similarity rate of distinct traces in the log	$\frac{1}{C_{\text{DT-\#}}(L) \cdot (C_{\text{DT-\#}}(L)-1)} \cdot \sum_{\sigma_1 \in L} \sum_{\sigma_2 \in L, \sigma_1 \neq \sigma_2} \left(1 - \frac{LD(\sigma_1,\sigma_2)}{\max(\lvert\sigma_1\rvert,\lvert\sigma_2\rvert)}\right)$, where $LD(\sigma_1,\sigma_2)$ is the Levensthein distance of two sequences σ_1, σ_2.	[8]
87.	$C_{\text{DA-min}}$	Minimum number of distinct activities per trace	$\min_{\sigma \in L} \{\lvert\{a \in A(L) \mid \exists i \in \{1,\ldots,\lvert\sigma\rvert\} : \sigma(i) = a\}\rvert\}$	[1]
88.	$C_{\text{DA-max}}$	Maximum number of distinct activities per trace	$\max_{\sigma \in L} \{\lvert\{a \in A(L) \mid \exists i \in \{1,\ldots,\lvert\sigma\rvert\} : \sigma(i) = a\}\rvert\}$	[1]
89.	$C_{\text{DA-avg}}$	Average number of distinct activities per trace	$\frac{\sum_{\sigma \in A(L)} \lvert\{a \in A(L) \mid \exists i \in \{1,\ldots,\lvert\sigma\rvert\} : \sigma(i) = a\}\rvert}{C_{\text{len}}(L)}$	[1, 8]
90.	$C_{\text{DA-dev}}$	Standard deviation of distinct activities per trace	$\sqrt{\frac{\sum_{\sigma \in L} (\lvert\{a \in A(L) \mid \exists i \in \{1,\ldots,\lvert\sigma\rvert\} : \sigma(i) = a\}\rvert - C_{\text{DA-avg}}(L))^2}{C_{\text{len}}(L)}}$	[1]

	Symbol	Description	Definition	Reference
91.	$C_{\text{DA-dens}}$	Density of distinct activities per trace	$\frac{C_{\text{DA-avg}}(L)}{C_{\text{TL-mean}}(L)}$	[8]
92.	$C_{\text{DA-ovrlp}}$	Average set-based non-overlap of traces	$1 - \frac{\sum_{\sigma_1,\sigma_2 \in L} L(\sigma_1) \cdot L(\sigma_2) \cdot \frac{\lvert A(\sigma_1) \cap A(\sigma_2) \rvert}{\lvert A(\sigma_1) \cup A(\sigma_2) \rvert}}{(\sum_{\sigma \in L} L(\sigma))^2}$	[1]
93.	$C_{\text{comp-fact}}$	Complexity factor of the event log	$\left(\ln\left(C_{\text{DT-\#}}(L)\right)^{((1-C_{\text{DT-sim}}(L))+C_{\text{DA-dens}}(L))} \cdot C_{\text{DA-avg}}(L)\right)$	[8]
94.	$C_{\text{s-T-div}}$	Simple trace diversity of the event log	$1 - \frac{C_{\text{DA-avg}}(L)}{C_{\text{var}}(L)}$	[8]
95.	$C_{\text{a-T-div}}$	Advanced trace diversity of the event log	$\frac{1}{C_{\text{DT-\#}}(L) \cdot (C_{\text{DT-\#}}(L)-1) \cdot C_{\text{TL-mean}}} \cdot \sum_{\sigma_1 \in L} \sum_{\sigma_2 \in L} LD(\sigma_1, \sigma_2)$, where $LD(\sigma_1, \sigma_2)$ is the Levensthein distance of two sequences σ_1, σ_2.	[8]
96.	$C_{\text{loop-\#}}$	Absolute number of traces with a loop	$\sum_{\sigma \in L, \lvert \mathscr{L}(\sigma) \rvert > 0} L(\sigma)$, where $\mathscr{L}(\sigma)$ is defined as the set $\{(i,\ell) \mid \exists a \in A(L) : \sigma(i) = \cdots = \sigma(i+\ell) = a \wedge i \in \{1, \ldots, \lvert\sigma\rvert - \ell\}\}$.	[8]
97.	$C_{\text{loop-rel-\#}}$	Relative number of traces with a loop	$\frac{C_{\text{loop-\#}}(L)}{C_{\text{len}}(L)}$	[8]
98.	$C_{\text{loop-avg-\#}}$	Average number of loops per trace	$\frac{\sum_{\sigma \in L, \lvert \mathscr{L}(\sigma) \rvert > 0} L(\sigma) \cdot \lvert \mathscr{L}(\sigma) \rvert}{C_{\text{len}}(L)}$, where $\mathscr{L}(\sigma)$ is defined as the set $\{(i,\ell) \mid \exists a \in A(L) : \sigma(i) = \cdots = \sigma(i+\ell) = a \wedge i \in \{1, \ldots, \lvert\sigma\rvert - \ell\}\}$.	[8]
99.	$C_{\text{loop-max-\#}}$	Maximum number of loops in a trace	$\max\{\lvert \mathscr{L}(\sigma) \rvert \mid \sigma \in L\}$, where $\mathscr{L}(\sigma)$ is defined as the set $\{(i,\ell) \mid \exists a \in A(L) : \sigma(i) = \cdots = \sigma(i+\ell) = a \wedge i \in \{1, \ldots, \lvert\sigma\rvert - \ell\}\}$.	[8]
100.	$C_{\text{loop-avg-size}}$	Average size of loops per trace	$\frac{\sum_{\sigma \in L, \lvert \mathscr{L}(\sigma) \rvert > 0} L(\sigma) \cdot \left(\sum_{(i,\ell) \in \mathscr{L}(\sigma)} \ell\right)}{C_{\text{len}}(L)}$, where $\mathscr{L}(\sigma)$ is defined as the set $\{(i,\ell) \mid \exists a \in A(L) : \sigma(i) = \cdots = \sigma(i+\ell) = a \wedge i \in \{1, \ldots, \lvert\sigma\rvert - \ell\}\}$.	[8]
101.	$C_{\text{loop-max-size}}$	Maximum size of loops in a trace	$\max\{\ell \mid \exists \sigma \in L : (i,\ell) \in \mathscr{L}(\sigma)\}$, where $\mathscr{L}(\sigma)$ is defined as the set $\{(i,\ell) \mid \exists a \in A(L) : \sigma(i) = \cdots = \sigma(i+\ell) = a \wedge i \in \{1, \ldots, \lvert\sigma\rvert - \ell\}\}$.	[8]
102.	$C_{\text{rep-\#}}$	Absolute number of traces with a repetition		[8]

	Symbol	Description	Definition	Reference
103.	$C_{\text{rep-rel-\#}}$	Relative number of traces with a repetition	$\frac{C_{\text{rep-rel-\#}}(L)}{C_{\text{len}}(L)}$	[8]
104.	$C_{\text{DFG-V}}$	Number of vertices in the directly follows graph	$\lvert V_{DFG(L)}\rvert$	[8]
105.	$C_{\text{DFG-E}}$	Number of edges in the directly follows graph	$\lvert E_{DFG(L)}\rvert$	[8]
106.	$C_{\text{DFG-cnc}}$	Coefficient of connectivity of the directly follows graph	$\frac{\lvert E_{DFG(L)}\rvert}{\lvert V_{DFG(L)}\rvert}$	[8]
107.	$C_{\text{DFG-adeg}}$	Average degree of nodes in the directly follows graph	$\frac{\sum_{v \in V_{DFG(L)}} deg(v)}{\lvert V_{DFG(L)}\rvert}$	[8]
108.	$C_{\text{DFG-mdeg}}$	Maximum degree of nodes in the directly follows graph	$\max\{deg(v) \mid v \in V_{DFG(L)}\}$	[8]
109.	$C_{\text{DFG-dens}}$	Density of the directly follows graph	$\frac{\lvert E_{DFG(L)}\rvert}{\lvert V_{DFG(L)}\rvert \cdot (\lvert V_{DFG(L)}\rvert - 1)}$	[8]
110.	$C_{\text{DFG-struct}}$	Structure of the directly follows graph	$1 - \frac{\lvert E_{DFG(L)}\rvert}{\lvert V_{DFG(L)}\rvert^2}$	[8]
111.	$C_{\text{DFG-CN}}$	Cyclomatic number of the directly follows graph	$\lvert E_{DFG(L)}\rvert - \lvert V_{DFG(L)}\rvert + 1$	[8]
112.	$C_{\text{DFG-diam}}$	Diameter of the directly follows graph	$\max\{n \mid \exists v_1, \ldots, v_n \in V_{DFG(L)} : (v_1, \ldots, v_n) \in \mathcal{P}_{DFG(L)}\}$, where $\mathcal{P}_{DFG(L)}$ is the set of simple paths in $DFG(L)$ that start in the start node and end in the final node.	[8]
113.	$C_{\text{DFG-sep-\#}}$	Number of cut vertices in the directly follows graph	$\{v \in V_{DFG(L)} \mid v \text{ is a cut-vertex of } DFG(L)\}$	[8]
114.	$C_{\text{DFG-sep-r}}$	Separability ratio of the directly follows graph	$\frac{C_{\text{DFG-sep-\#}}(L)}{\lvert V_{DFG(L)}\rvert}$	[8]
115.	$C_{\text{DFG-seq-r}}$	Sequentiality ratio of the directly follows graph	$\{(u,v) \in E_{DFG(L)} \mid u, v \in \mathcal{S}_{DFG(L)}\}$, where $\mathcal{S}_{DFG(L)}$ is the set of sequential nodes, i.e., $\mathcal{S}_{DFG(L)} = \{x \in V_{DFG(L)} \mid \lvert\{y \mid (x,y) \in E_{DFG(L)}\}\rvert \leq 1 \wedge \lvert\{y \mid (y,x) \in E_{DFG(L)}\}\rvert \leq 1\}$.	[8]

	Symbol	Description	Definition	Reference
116.	$C_{\text{DFG-cyc}}$	Cyclicity of the directly follows graph	$\frac{\lvert\{v \in V_{DFG(L)} \mid v \text{ lies on a cycle in } DFG(L)\}\rvert}{\lvert V_{DFG(L)}\rvert}$	[8]
117.	$C_{\text{DFG-P-comp}}$	Simple path complexity of the directly follows graph	$\lvert\{(v_1,\ldots,v_n) \mid v_1,\ldots,v_n \in dfgV \wedge (v_1,\ldots,v_n) \in \mathcal{P}_{DFG(L)}\}\rvert$, where $\mathcal{P}_{DFG(L)}$ is the set of simple paths in $DFG(L)$ that start in the start node and end in the final node.	[8]
118.	C_{succ}	Number of $>_L$-relations between activities	$\lvert >_L \rvert$	[1]
119.	C_{ties}	Number of $\rightarrow_L$-relations between activities	$\lvert \rightarrow_L \rvert$	[2]
120.	C_{struct}	Inverse ratio of directly follows relations	$1 - \frac{\lvert\rightarrow_L\rvert}{\lvert A(L)\rvert^2}$	[4]
121.	C_{affinity}	Average number of traces' common neighborhoods	$\frac{\left(\sum_{\sigma_1,\sigma_2 \in L} L(\sigma_1)\cdot L(\sigma_2)\cdot aff(\sigma_1,\sigma_2)\right) - \sum_{\sigma\in L,\lvert\sigma\rvert>1} L(\sigma)}{\left(\sum_{\sigma\in L} L(\sigma)\right)\cdot\left(\left(\sum_{\sigma\in L} L(\sigma)\right)-1\right)}$, where $aff(\sigma_1,\sigma_2) = \frac{\lvert\mathcal{N}(\sigma_1)\cap\mathcal{N}(\sigma_2)\rvert}{\lvert\mathcal{N}(\sigma_1)\cup\mathcal{N}(\sigma_2)\rvert}$ and $\mathcal{N}(\sigma) = \{(a,b) \mid \exists i \in \{1,\ldots,\lvert\sigma\rvert-1\} : \sigma(i) = a \wedge \sigma(i+1) = b\}$.	[4]
122.	C_{LZ}	Lempel-Ziv complexity of a sequentialization of the log	$LZ\left(\prod_{\sigma\in L} \sigma^{L(\sigma)}\right)$, where $LZ(w)$ denotes the Lempel-Ziv complexity [5] of a sequence w.	[7]
123.	$C_{\text{dev-R}}$	Standard deviation from a completely random log	$1 - \sqrt{\sum_{(e_1,e_2)\in A(L)\times A(L)} \left(\frac{\left\lvert n^L_{e_1\rightarrow e_2} - \frac{\sum_{\sigma\in L} L(\sigma)\cdot(\lvert\sigma\rvert-1)}{\lvert A(L)\rvert^2}\right\rvert}{\sum_{\sigma\in L} L(\sigma)\cdot(\lvert\sigma\rvert-1)}\right)}$, where $n^L_{e_1\rightarrow e_2}$ is the number of times event e_1 is directly followed by event e_2 in L.	[7]
124.	$C_{\text{avg-dist}}$	Average Levensthein distance between the traces	$\frac{\sum_{\sigma_1,\sigma_2\in L} LD(\sigma_1,\sigma_2)}{\left(\sum_{\sigma\in L} L(\sigma)\right)\cdot\left(\left(\sum_{\sigma\in L} L(\sigma)\right)-1\right)}$, where $LD(\sigma_1,\sigma_2)$ is the Levensthein distance of two sequences σ_1,σ_2.	[7]

	Symbol	Description	Definition	Reference
125.	$C_{\text{ENT-trace}}$	Entropy of trace frequencies in the log	$-\sum_{\sigma \in L} \frac{L(\sigma)}{C_{\text{len}}(L)} \cdot \log_2 \left(\frac{L(\sigma)}{C_{\text{len}}(L)} \right)$	[3, 6]
126.	$C_{\text{ENT-pref}}$	Entropy of frequencies of trace-prefixes in the log	$-\sum_{\omega \in A(L)^*} \frac{\sum_{\sigma \in Ex_L(\omega)} L(\sigma)}{C_{\text{mag}}(L)} \cdot \log_2 \left(\frac{\sum_{\sigma \in Ex_L(\omega)} L(\sigma)}{C_{\text{mag}}(L)} \right)$, where $Ex_L(\omega) = \{\sigma \in L \mid \exists i \in \{1, \ldots, \lvert\sigma\rvert\} : \omega = \sigma(1) \ldots \sigma(i)\}$.	[3, 6]
127.	$C_{\text{ENT-pref-F}}$	Entropy of trace-prefixes in the log's support	$-\sum_{\omega \in A(L)^*} \frac{\lvert Ex_L(\omega)\rvert}{\sum_{\sigma \in L} \lvert\sigma\rvert} \cdot \log_2 \left(\frac{\lvert Ex_L(\omega)\rvert}{\sum_{\sigma \in L} \lvert\sigma\rvert} \right)$, where $Ex_L(\omega) = \{\sigma \in L \mid \exists i \in \{1, \ldots, \lvert\sigma\rvert\} : \omega = \sigma(1) \ldots \sigma(i)\}$.	[3, 6]
128.	$C_{\text{ENT-g-block}}$	Global block entropy of the log	$-\sum_{\omega \in A(L)^*} \frac{\sum_{\sigma \in Bl_L(\omega)} L(\sigma)}{\sum_{\sigma \in L} \frac{1}{2} L(\sigma) \cdot (\lvert\sigma\rvert \cdot (\lvert\sigma\rvert - 1))} \cdot \log_2 \left(\frac{\sum_{\sigma \in Bl_L(\omega)} L(\sigma)}{\sum_{\sigma \in L} \frac{1}{2} L(\sigma) \cdot (\lvert\sigma\rvert \cdot (\lvert\sigma\rvert - 1))} \right)$, where $Bl_L(\omega) = \{\sigma \in L \mid \exists i \in \{1, \ldots, \lvert\sigma\rvert - \lvert\omega\rvert\} : \omega = \sigma(i) \ldots \sigma(i + \lvert\omega\rvert)\}$	[3, 6]
129.	$C_{\text{ENT-g-block-F}}$	Global block entropy of the log's support	$-\sum_{\omega \in A(L)^*} \frac{\lvert Bl_L(\omega)\rvert}{\sum_{\sigma \in L} \frac{1}{2} (\lvert\sigma\rvert \cdot (\lvert\sigma\rvert - 1))} \cdot \log_2 \left(\frac{\lvert Bl_L(\omega)\rvert}{\sum_{\sigma \in L} \frac{1}{2} (\lvert\sigma\rvert \cdot (\lvert\sigma\rvert - 1))} \right)$, where $Bl_L(\omega) = \{\sigma \in L \mid \exists i \in \{1, \ldots, \lvert\sigma\rvert - \lvert\omega\rvert\} : \omega = \sigma(i) \ldots \sigma(i + \lvert\omega\rvert)\}$	[3, 6]
130.	$C_{\text{ENT-LZ}}$	Lempel-Ziv entropy rate of the log	$-\sum_{\sigma \in L} \frac{LZ\left(\prod_{\sigma \in L} \sigma^{L(\sigma)}\right) \cdot \sum_{\sigma \in L} L(\sigma) \cdot \lvert\sigma\rvert}{C_{\text{mag}}(L)} \cdot \log_2 \left(\frac{LZ\left(\prod_{\sigma \in L} \sigma^{L(\sigma)}\right) \cdot \sum_{\sigma \in L} L(\sigma) \cdot \lvert\sigma\rvert}{C_{\text{mag}}(L)} \right)$, where $LZ(w)$ denotes the Lempel-Ziv complexity [5] of a sequence w.	[3, 6]
131.	$C_{\text{ENT-LZ-F}}$	Lempel-Ziv entropy rate of the log's support	$-\sum_{\sigma \in L} \frac{LZ\left(\prod_{\sigma \in L} \sigma\right) \cdot \sum_{\sigma \in L} \lvert\sigma\rvert}{\sum_{\sigma \in L} \lvert\sigma\rvert} \cdot \log_2 \left(\frac{LZ\left(\prod_{\sigma \in L} \sigma\right) \cdot \sum_{\sigma \in L} \lvert\sigma\rvert}{\sum_{\sigma \in L} \lvert\sigma\rvert} \right)$, where $LZ(w)$ denotes the Lempel-Ziv complexity [5] of a sequence w.	[3, 6]
132.	$C_{\text{ENT-k-block-d1}}$	Difference-based 1-block entropy of the log	$ent^2_{Bl}(L) - ent^1_{Bl}(L)$, where $ent^k_{Bl}(L) = -\sum_{\omega in A(L)^*} \frac{\sum_{\sigma \in Bl^k_L(\omega)} L(\sigma)}{\sum_{\sigma \in L} L(\sigma) \cdot \max(0, \lvert\sigma\rvert - k + 1)} \cdot \log_2 \left(\frac{\sum_{\sigma \in Bl^k_L(\omega)} L(\sigma)}{\sum_{\sigma \in L} L(\sigma) \cdot \max(0, \lvert\sigma\rvert - k + 1)} \right)$ and $Bl^k_L(\omega) = \{\sigma \in L \mid \exists i \in \{1, \ldots, \lvert\sigma\rvert - k\} : \omega = \sigma(i) \ldots \sigma(i + k)\}$.	[3, 6]
133.	$C_{\text{ENT-k-block-d3}}$	Difference-based 3-block entropy of the log	$ent^4_{Bl}(L) - ent^3_{Bl}(L)$, where $ent^k_{Bl}(L) = -\sum_{\omega in A(L)^*} \frac{\sum_{\sigma \in Bl^k_L(\omega)} L(\sigma)}{\sum_{\sigma \in L} L(\sigma) \cdot \max(0, \lvert\sigma\rvert - k + 1)} \cdot \log_2 \left(\frac{\sum_{\sigma \in Bl^k_L(\omega)} L(\sigma)}{\sum_{\sigma \in L} L(\sigma) \cdot \max(0, \lvert\sigma\rvert - k + 1)} \right)$ and $Bl^k_L(\omega) = \{\sigma \in L \mid \exists i \in \{1, \ldots, \lvert\sigma\rvert - k\} : \omega = \sigma(i) \ldots \sigma(i + k)\}$.	[3, 6]

	Symbol	Description	Definition	Reference
134.	$C_{\text{ENT-k-block-d5}}$	Difference-based 5-block entropy of the log	$ent^6_{Bl}(L) - ent^5_{Bl}(L)$, where $ent^k_{Bl}(L) = -\sum_{\omega in A(L)^*} \frac{\sum_{\sigma \in Bl^k_L(\omega)} L(\sigma)}{\sum_{\sigma \in L} L(\sigma) \cdot \max(0, \lvert\sigma\rvert - k + 1)} \cdot \log_2 \left(\frac{\sum_{\sigma \in Bl^k_L(\omega)} L(\sigma)}{\sum_{\sigma \in L} L(\sigma) \cdot \max(0, \lvert\sigma\rvert - k + 1)} \right)$ and $Bl^k_L(\omega) = \{\sigma \in L \mid \exists i \in \{1, \ldots, \lvert\sigma\rvert - k\} : \omega = \sigma(i) \ldots \sigma(i+k)\}$.	[3, 6]
135.	$C_{\text{ENT-k-block-r1}}$	Ratio-based 1-block entropy of the log	$ent^1_{Bl}(L)$, where $ent^k_{Bl}(L) = -\sum_{\omega in A(L)^*} \frac{\sum_{\sigma \in Bl^k_L(\omega)} L(\sigma)}{\sum_{\sigma \in L} L(\sigma) \cdot \max(0, \lvert\sigma\rvert - k + 1)} \cdot \log_2 \left(\frac{\sum_{\sigma \in Bl^k_L(\omega)} L(\sigma)}{\sum_{\sigma \in L} L(\sigma) \cdot \max(0, \lvert\sigma\rvert - k + 1)} \right)$ and $Bl^k_L(\omega) = \{\sigma \in L \mid \exists i \in \{1, \ldots, \lvert\sigma\rvert - k\} : \omega = \sigma(i) \ldots \sigma(i+k)\}$.	[3, 6]
136.	$C_{\text{ENT-k-block-r3}}$	Ratio-based 3-block entropy of the log	$\frac{1}{3} \cdot ent^3_{Bl}(L)$, where $ent^k_{Bl}(L) = -\sum_{\omega in A(L)^*} \frac{\sum_{\sigma \in Bl^k_L(\omega)} L(\sigma)}{\sum_{\sigma \in L} L(\sigma) \cdot \max(0, \lvert\sigma\rvert - k + 1)} \cdot \log_2 \left(\frac{\sum_{\sigma \in Bl^k_L(\omega)} L(\sigma)}{\sum_{\sigma \in L} L(\sigma) \cdot \max(0, \lvert\sigma\rvert - k + 1)} \right)$ and $Bl^k_L(\omega) = \{\sigma \in L \mid \exists i \in \{1, \ldots, \lvert\sigma\rvert - k\} : \omega = \sigma(i) \ldots \sigma(i+k)\}$.	[3, 6]
137.	$C_{\text{ENT-k-block-r5}}$	Ratio-based 5-block entropy of the log	$\frac{1}{5} \cdot ent^5_{Bl}(L)$, where $ent^k_{Bl}(L) = -\sum_{\omega in A(L)^*} \frac{\sum_{\sigma \in Bl^k_L(\omega)} L(\sigma)}{\sum_{\sigma \in L} L(\sigma) \cdot \max(0, \lvert\sigma\rvert - k + 1)} \cdot \log_2 \left(\frac{\sum_{\sigma \in Bl^k_L(\omega)} L(\sigma)}{\sum_{\sigma \in L} L(\sigma) \cdot \max(0, \lvert\sigma\rvert - k + 1)} \right)$ and $Bl^k_L(\omega) = \{\sigma \in L \mid \exists i \in \{1, \ldots, \lvert\sigma\rvert - k\} : \omega = \sigma(i) \ldots \sigma(i+k)\}$.	[3, 6]
138.	$C_{\text{ENT-knn-3}}$	Third nearest neighbor entropy	$\frac{1}{C_{\text{len}}(L)} \cdot \sum_{\sigma \in L} d_3(\sigma) + \log_2 \left(\frac{\sqrt{\pi}}{(1.5)!} \right) + \gamma - \sum_{x=1}^{2} + \log_2 \left(C_{\text{len}}(L) \right)$, where $\gamma \approx 0.5772$ is Euler's constant and $d_k(\sigma) = LD(\sigma, \omega_k)$, where $(\omega_1, \ldots, \omega_{C_{\text{len}}(L)})$ is the log L, sorted in ascending order w.r.t. $LD(\sigma, w_i)$ for $i \in \{1, \ldots, C_{\text{len}}(L)\}$, and LD denotes the Levensthein distance between two sequences.	[3, 6]
139.	$C_{\text{ENT-knn-5}}$	Fifth nearest neighbor entropy	$\frac{1}{C_{\text{len}}(L)} \cdot \sum_{\sigma \in L} d_5(\sigma) + \log_2 \left(\frac{\sqrt{\pi}}{(1.5)!} \right) + \gamma - \sum_{x=1}^{4} + \log_2 \left(C_{\text{len}}(L) \right)$, where $\gamma \approx 0.5772$ is Euler's constant and $d_k(\sigma) = LD(\sigma, \omega_k)$, where $(\omega_1, \ldots, \omega_{C_{\text{len}}(L)})$ is the log L, sorted in ascending order w.r.t. $LD(\sigma, w_i)$ for $i \in \{1, \ldots, C_{\text{len}}(L)\}$, and LD denotes the Levensthein distance between two sequences.	[3, 6]

	Symbol	Description	Definition	Reference
140.	$C_{\text{ENT-knn-7}}$	Seventh nearest neighbor entropy	$\frac{1}{C_{\text{len}}(L)} \cdot \sum_{\sigma \in L} d_7(\sigma) + \log_2\left(\frac{\sqrt{\pi}}{(1.5)!}\right) + \gamma - \sum_{x=1}^{6} + \log_2\left(C_{\text{len}}(L)\right)$, where $\gamma \approx 0.5772$ is Euler's constant and $d_k(\sigma) = LD(\sigma, \omega_k)$, where $(\omega_1, \ldots, \omega_{C_{\text{len}}(L)})$ is the log L, sorted in ascending order w.r.t. $LD(\sigma, w_i)$ for $i \in \{1, \ldots, C_{\text{len}}(L)\}$, and LD denotes the Levensthein distance between two sequences.	[3, 6]
141.	$C_{\text{epa-var}}$	Variant entropy of the log's prefix automaton	...	[3, 6]
142.	$C_{\text{epa-n-var}}$	Normalized variant entropy of the log's prefix automaton	...	[2, 6]
143.	$C_{\text{epa-seq}}$	Sequence entropy of the log's prefix automaton	...	[2, 6]
144.	$C_{\text{epa-n-seq}}$	Normalized sequence entropy of the log's prefix automaton	...	[2, 6]
145.	$C_{\text{epa-seq-L}}$	...	...	[6]
146.	$C_{\text{epa-n-seq-L}}$	...	...	[6]
147.	$C_{\text{epa-seq-E}}$	...	...	[6]
148.	$C_{\text{epa-n-seq-E}}$	...	...	[6]

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